
Automotive SoC Motor Driver

1.0 GENERAL PURPOSE TIMER UNIT (GTU)

The general purpose timer unit hosts eight 16-bit timer-counters. Each timer is fully configurable to work independently of the remaining timers. Each timer can function as an event counter operating on an input signal or as a dedicated timer running on the input clock.

1.1 Features

- 16-bit counter
- 16 selectable input source signals
- Programmable clock pre-scaler
- Ten modes of operation:
 - o High-time Measurement
 - o Low-time Measurement
 - o Period Measurement: from rising edge to rising edge
 - o Period Measurement: from falling edge to falling edge
 - o Event Counter: rising edges
 - o Event Counter: falling edges
 - o Event Counter: rising or falling edges
 - o 4 Fixed Duty-Cycle Trigger Generation (FDCTG) on GPO
 - o Start/Stop timer mode
 - o 4 Variable Duty-Cycle Trigger Generation (VDCTG) on GPO
- Counter overflow, data valid and overrun status bits
- Automatic restart of timers when threshold value is reached, or overflow is flagged
- Independent interrupt request for each timer channel
- Configurable interrupt sources

1.2 Introduction

The GTU module provides 8 timer channels that can be used to keep track of occurring events on any input signal. The timer can be configured to count the number of events detected until a certain programmable threshold is reached. It can also provide period measurements on input signals and report to the host MCU the detected period in units of the timer clock. Finally, the timers can be configured as general timers running on the input clock which can be controlled via software. It is also possible to generate 4 trigger signals with programmable duty cycle and period on GPO using timer module.

1.3 Functional Description

Each timer is independent of the remaining timers; meaning it could be configured separately to operate in any mode of operation regardless of the remaining timers' configuration. Each timer has configurable interrupt triggers that are monitored by an interrupt system which generates a corresponding interrupt request (Tx_IRQ) to the Cortex M4's Nested Vectored Interrupt Controller (NVIC). The module also hosts a clock prescaler function to add extra flexibility and extend the range of the measured signals. Each timer can output a trigger signal on general purpose output (GPO) pins when configured to operate in FDCTG or VDCTG modes. The trigger signal is also output to data acquisition unit (DAU) and current acquisition unit (CAU) to sample the desired signal, such as supply voltage, at a programmed rate. For further information check CAU and DAU documentation. Timer unit has a dedicated control register to configure the timer's mode of operation, pre-scaler settings, interrupt trigger source and input signal source. Each timer also features a threshold register to program the threshold value of timers operating in counter mode. A status register is included for each timer to monitor its state, and provide information regarding any overflow and data validity. Lastly, a result register is included which shows the value of the counter at any given time. All registers are accessed through the AMBA APB bus interface. Figure 1-1 shows the general block diagram of the general purpose timer 1. The same structure is followed for the remaining 7 other timer channels. It should be noted only timers 1-4 Trigger outputs (TX_TRIGGER) are connected to GPIO, CAU and DAU module.

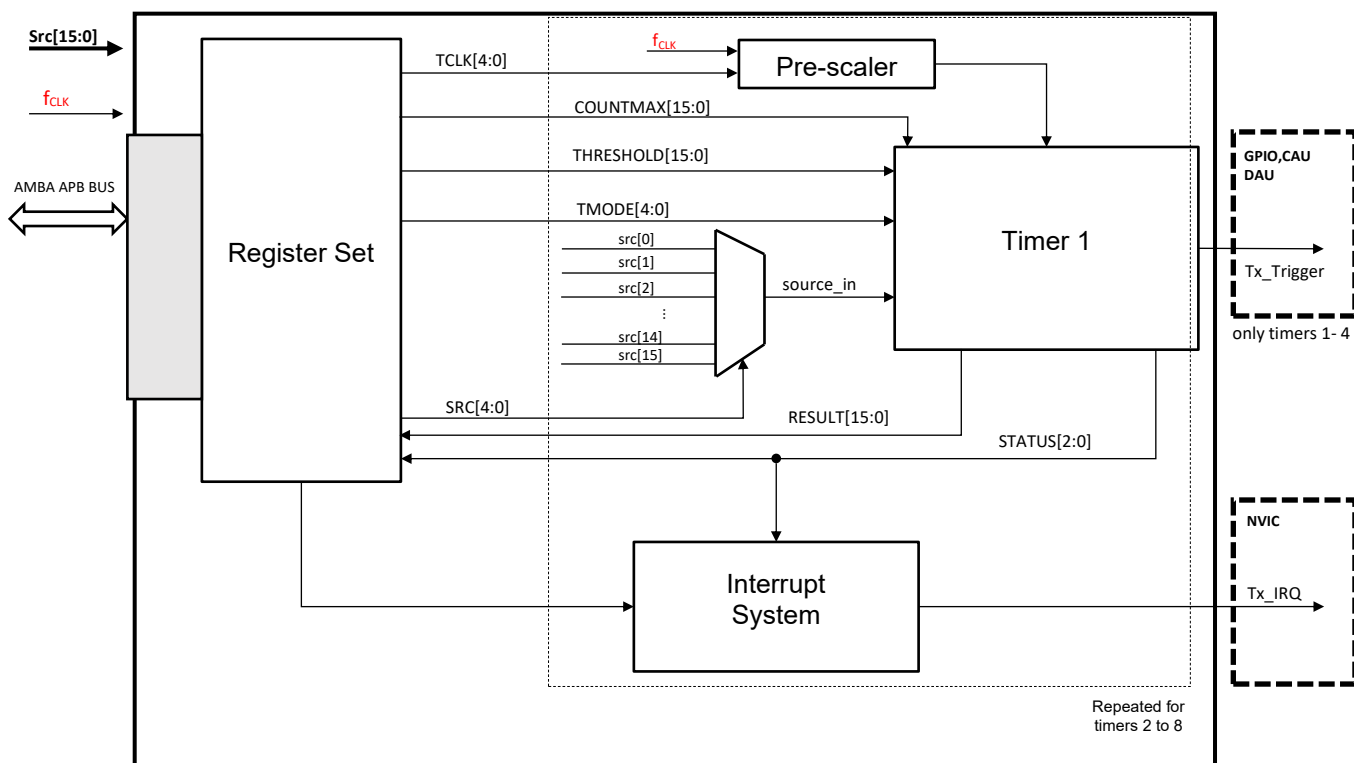


Figure 1-1: General block diagram of the general-purpose timer 1

1.4 Modes of operation

The timer-counter units have 10 different modes of operation that can be selected through the TMODE bits in the control register. This section explains each mode of operation. The following are the available modes of operation:

1. High-time Measurement mode
2. Low-time Measurement mode
3. Period Measurement mode – Rising edge to rising edge
4. Period Measurement mode – Falling edge to falling edge
5. Event Counter mode – Rising edges
6. Event Counter mode – Falling edges
7. Event Counter mode – Rising or falling edges
8. Fixed Duty-Cycle Trigger Generation (FDCTG)
9. Timer mode – Start/Stop timer
10. Variable Duty-Cycle Trigger Generation (VDCTG)

1.4.1 High-time Measurement mode

In high-time measurement mode, selected by setting TMODE to '0' in the control register, the timer-counter is configured to measure the duration of a level high signal detected on the selected input source. The measurement cycle starts on the detection of a rising edge on the selected input source, and finishes on the detection of a falling edge on the selected input source. During the measurement cycle, the counter counts up with the selected sample rate, which is configured by setting TCLK in the control register. This mode starts once TIMER_EN in the control register is set to '1'.

When the falling edge is detected on the selected input signal, the result register is updated with the counter value, and the DVL bit is set to '1' in the status register indicating that a measurement is complete and data is valid and ready to be read. An IRQ is triggered if the option is enabled by IEN_DVL bit in the control register. The internal logic is automatically restarted and becomes ready to start a new measurement cycle on the next rising edge of the input source signal. Figure 1-2 shows the operation of the counter in high-time measurement mode.

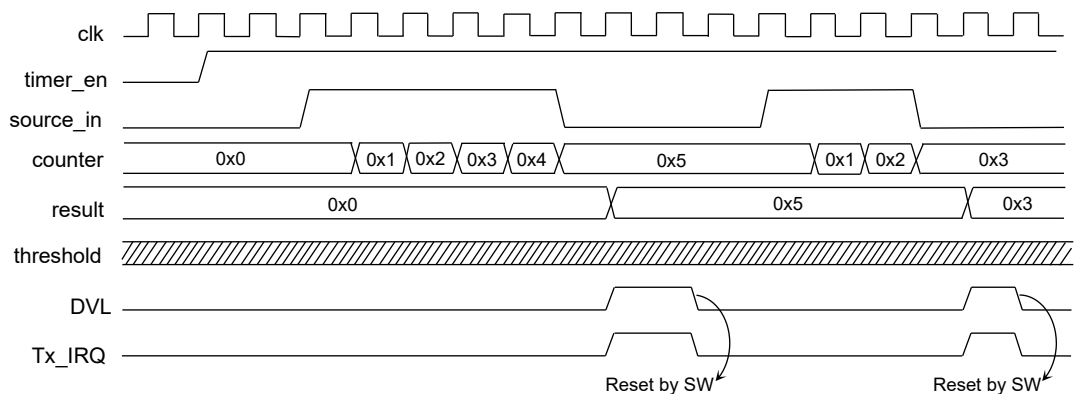


Figure 1-2: Counter's operation in high-time measurement mode

1.4.2 Low-time Measurement mode

In low-time measurement mode, selected by setting TMODE to '1' in the control register, the timer-counter is configured to measure the duration of a low level signal detected on the selected input source. The measurement cycle starts on the detection of a falling edge on the selected input source, and finishes on the detection of a rising edge on the selected input source. During the measurement cycle, the counter counts up with the selected sample rate, which is configured by setting TCLK in the control register. This mode starts once TIMER_EN in the control register is set to '1'.

When the rising edge is detected on the selected input signal, the result register is updated with the counter value, and the DVL bit is set to '1' in the status register indicating that a measurement is complete and data is valid and ready to be read. An IRQ is triggered if the option is enabled by IEN_DVL bit in the control register. The internal logic is automatically restarted and becomes ready to start a new measurement cycle on the next falling edge of input source signal. Figure 1-3 shows the operation of the counter in low-time measurement mode.

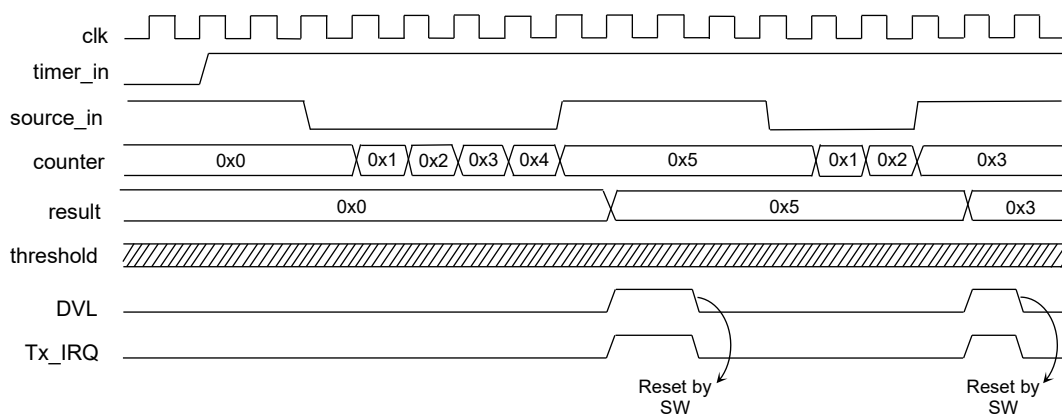


Figure 1-3: Counter's operation in low-time measurement mode

1.4.3 Period Measurement mode – Rising edge to rising edge

In period measurement mode, the duration between two consecutive rising edges can be calculated. This mode can be selected by setting TMODE to '2' in the control register. The measurement cycle starts with the detection of a rising edge on the selected input source signal, and it ends with the detection of the next rising edge on the selected input source signal. During the measurement cycle the counter is incremented with the selected source sample rate, which is configured by setting the TCLK bits in the control register. This mode starts once TIMER_EN in the control register is set to '1'.

At the end of the measurement cycle (ie. at every rising edge on the input source), the result register is updated with the final value of the counter and the DVL bit in the status register is set to '1', and the measurement of the following period starts automatically where the counter starts counting from '1'. An IRQ is triggered if the option is enabled by IEN_DVL bit in the control register. Figure 1-5 shows the operation of the counter in period measurement mode (rising edge to rising edge).

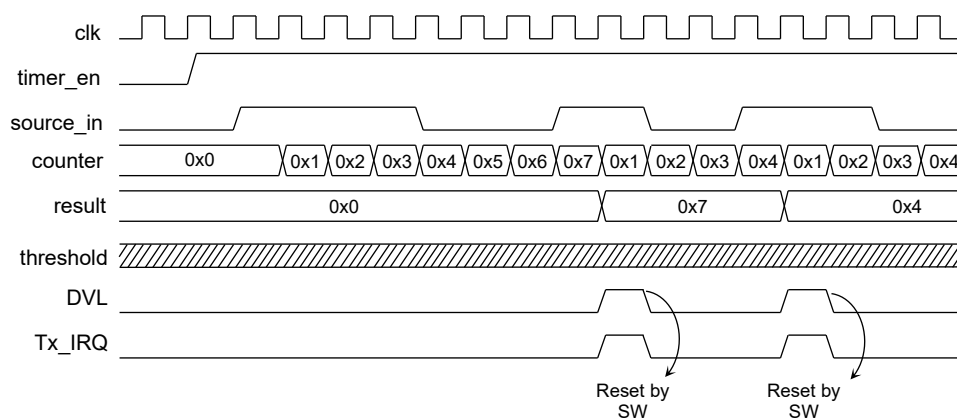


Figure 1-4: Counter's operation in period measurement mode (rising edge to rising edge)

1.4.4 Period Measurement mode – Falling edge to falling edge

In period measurement mode, the duration between two consecutive falling edges can be calculated. This mode can be selected by setting TMODE to '3' in the control register. The measurement cycle starts with the detection of a falling edge on the selected input source signal, and it ends with the detection of the next falling edge on the selected input source signal. During the measurement cycle, the counter is incremented with the selected source sample rate, which is configured by setting the TCLK bits in the control register. This mode starts once TIMER_EN in the control register is set to '1'.

At the end of the measurement cycle (ie. at every falling edge on the input source), the result register is updated with the final value of the counter and the DVL bit in the status register is set to '1', and the measurement of the following period starts automatically where the counter starts counting from 1. An IRQ is triggered if the option is enabled by IEN_DVL bit in the control register. Figure 1-5 shows the operation of the counter in period measurement mode (falling edge to falling edge).

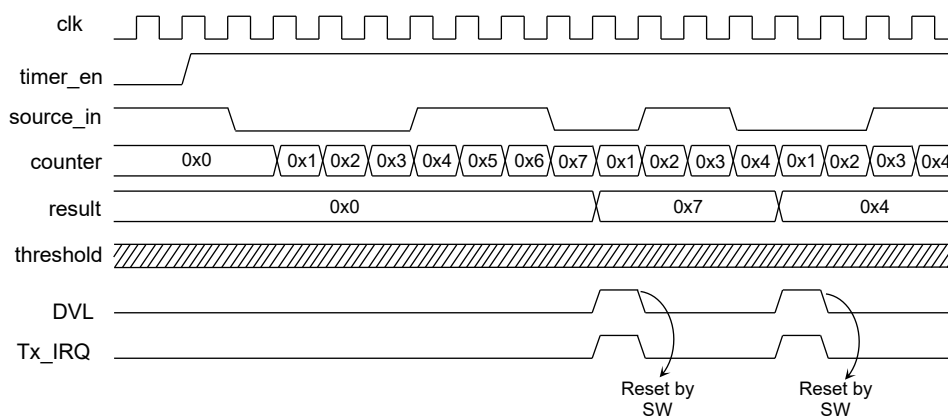


Figure 1-5: Counter's operation in period measurement mode (falling edge to falling edge)

1.4.5 Event Counter mode – Rising edges

In this mode, the counter is configured to increment on every detection of a rising edge on the input source signal. This mode is selected by setting TMODE to '4' in the control register. The counter starts counting from 1 until it reaches the threshold value defined by the threshold register. This mode starts once TIMER_EN in the control register is set to '1'.

The result register is continuously updated with the current value of the counter. The measurement cycle ends when the counter value reaches the threshold value. The DVL bit in the status register is set to '1' and an IRQ is triggered if the option is enabled using the IEN_DVL bit in the control register. The counter will automatically restart and will be ready to count the next detected rising edge. Figure 1-6 shows the operation of the counter in event counter mode (rising edges).

In the special case where the threshold value is 0, no event counting will take place. The status register, event counter and result register are all reset to 0.

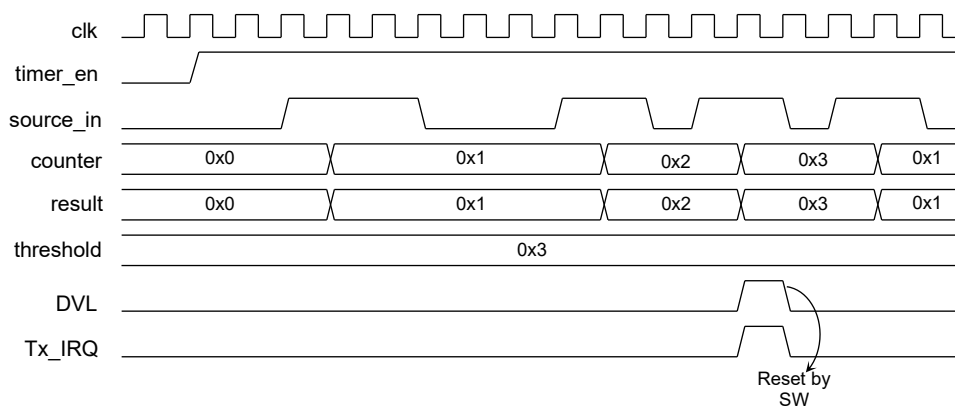


Figure 1-6: Counter's operation in event counter mode (rising edges)

1.4.6 Event Counter mode – Falling edges

In this mode, the counter is configured to increment on every detection of a falling edge on the input source signal. This mode is selected by setting TMODE to '5' in the control register. The counter starts counting from 1 until it reaches the threshold value defined by the threshold register. This mode starts once TIMER_EN in the control register is set to '1'.

The result register is continuously updated with the current value of the counter. The measurement cycle ends when the counter value reaches the threshold value. The DVL bit in the status register is set to '1' and an IRQ is triggered if the option is enabled using the IEN_DVL bit in the control register. The counter will automatically restart and will be ready to count the next detected falling edge. Figure 1-7 shows the operation of the counter in event counter mode (falling edges).

In the special case where the threshold value is 0, no event counting will take place. The status register, event counter and result register are all reset to 0.

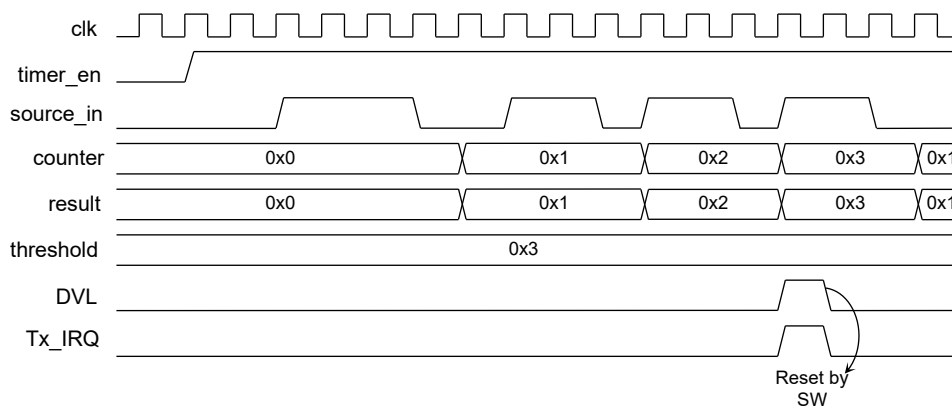


Figure 1-7: Counter's operation in event counter mode (falling edges)

1.4.7 Event Counter mode – Rising or falling edges

In this mode, the counter is configured to increment on every detection of a rising or falling edge on the input source signal. This mode is selected by setting TMODE = 0x6 in the control register. The counter starts counting from 1 until it reaches the threshold value defined by the threshold register. This mode starts once TIMER_EN in the control register is set to '1'.

The result register is continuously updated with the current value of the counter. The measurement cycle ends when the counter value reaches the threshold value set by threshold register. The DVL bit in the status register is set to '1' and an IRQ is triggered if the option is enabled using the IEN_DVL bit in the control register. The counter will automatically restart and will be ready to count the next detected edge. Figure 1-8 shows the operation of the counter in event counter mode (rising or falling edges).

In the special case where the threshold value is set to 0, no event counting will take place. The status register, event counter and result register are all reset to 0.

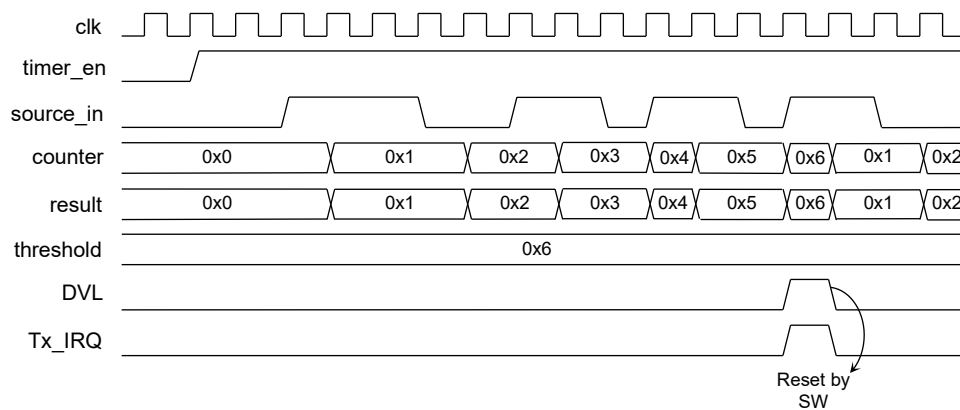


Figure 1-8: Counter's operation in event counter mode (rising or falling edges)

1.4.8 Fixed Duty-Cycle Trigger Generation (FDCTG)

In this mode, the timer is incremented with the clock rate specified by the TCLK bits in the control register. This mode is selected by setting TMODE to '7' in the control register. The timer starts from the value 1 and counts up to the threshold value defined by the threshold register. Once the threshold value is reached, the timer automatically resets and starts counting up from 1 again until it reaches the threshold value and this loop is repeated. The input source signal is not used in this mode. The result register is continuously updated with the counter value. This mode starts once TIMER_EN in the control register is set to '1'.

The timer will generate an output signal TX_TRIGGER in one measurement cycle as follows. The output of TX_TRIGGER is set to '1' when the timer value reaches half the threshold value (floor rounded). The output is set to '0' when the timer value reaches the threshold value, which is the end of the measurement cycle. Once the measurement cycle is complete, the DVL flag is set to '1' and an IRQ is triggered if the option is enabled by the IEN_DVL bit in the control register. Figure 1-9 shows the operation of the counter in fixed duty-cycle trigger generation mode.

The generated TX_TRIGGER signal has 50% duty cycle when the threshold is an even number and approximately 50% duty cycle when the threshold is an odd number (due to the rounding operation). The period of the TX_TRIGGER signal is specified by the following equation:

$$TX_TRIGGER_{period} \text{ (seconds)} = THRESHOLD \times t_{CLK} \times \text{prescaler}$$

$$\text{where: } t_{CLK} = \frac{1}{f_{CLK} \text{ (Hz)}}$$

In the special cases where the threshold value is equal to zero or one, the timer does not run and will be reset to 0. The result register reads 0 and the TX_TRIGGER output is constantly low.

The TX_TRIGGER signal for timers 5, 6 and 6 are also fed back to the timer unit and can be used in other modes of operation. For example if timer 5 is set to FDCTG mode, timer 2 can be set to event counter mode, set the source signal to T5_TRIGGER and count the number of rising edge triggers for T5_TRIGGER if desired.

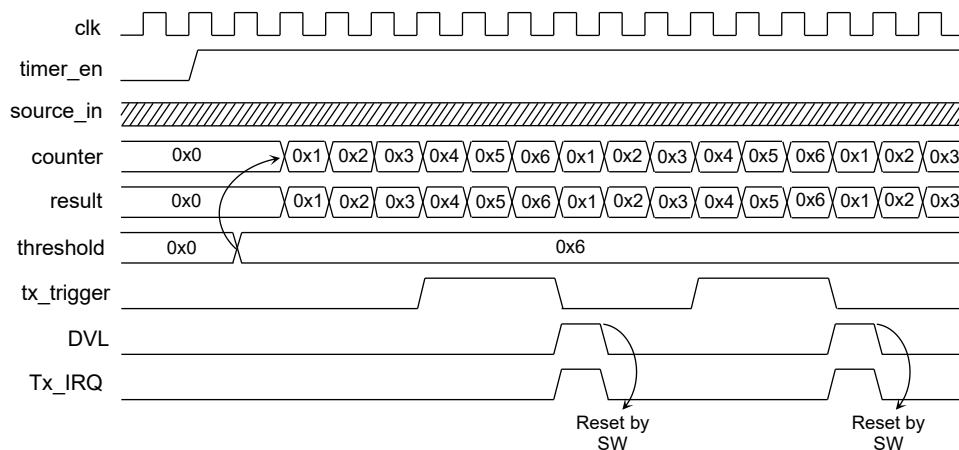


Figure 1-9: Counter's operation in Fixed Duty-Cycle Trigger Generation (FDCTG)

1.4.9 Timer mode – Start/Stop timer

In this mode, the timer acts as a free-running timer controlled solely by the user software. This mode is selected by setting `TMODE = 0x8` in the control register. The timer operation is controlled via the `RUN_NSTOP` bit in the StartStop register. When the `RUN_NSTOP` bit is set to '1', the timer starts incrementing by one with the clock rate specified by the `TCLK` bits in the control register.

When the `RUN_NSTOP` bit is cleared to '0' the timer freezes and stops incrementing, keeping its current state. If the `RUN_NSTOP` bit is set once again to '1' the timer continues incrementing from the previously stored state.

If the timer continue to increment until it reaches its maximum value (0xFFFF), the timer will roll over to 0, continue to increment, and the `OVF` flag is set in the status register to indicate that an overflow has occurred. An `IRQ` is triggered on the overflow event if the option is enabled via the `IEN_OVF` bit in the control register. At any time, the `result` displays the timer value.

In order to reset the timer, the `TIMER_EN` bit in the control register must be reset to '0' and then re-enabled again by setting it to '1'. This will clear the timer state and reset the timer value. This mode starts once `TIMER_EN` in the control register is set to '1'.

In this mode, the threshold value defined by the threshold register is irrelevant. Input source signals are not used in this mode of operation. Figure 1-10 shows the operation of the timer in start/stop timer mode.

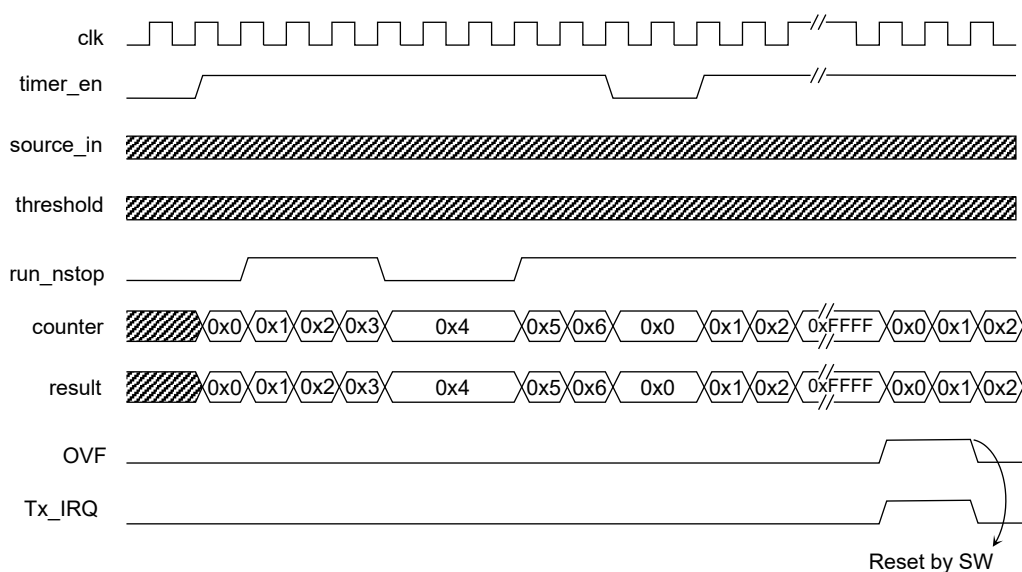


Figure 1-10: Timer operation in start/stop timer mode

1.4.10 Variable Duty-Cycle Trigger Generation (VDCTG)

In this mode, the timer is incremented with the clock rate specified by the TCLK bits in the control register. This mode is selected by setting TMODE to '9' in the control register. The timer starts from the value 1 and counts up to the COUNTMAX value defined by the countmax register. Once the COUNTMAX value is reached, the timer automatically resets and starts counting up from 1 again until it reaches the COUNTMAX value and this loop is repeated. The input source signal is not used in this mode. The result register is continuously updated with the counter value. This mode starts once TIMER_EN in the control register is set to '1'.

The timer will generate an output signal TX_TRIGGER in one measurement cycle as follows. The output of TX_TRIGGER is set high until the timer reaches to a THRESHOLD value defined by threshold register. The output is then set low when the timer value reaches the COUNTMAX value in countmax register. The TX_TRIGGER signal remains low until the timer reaches to COUNTMAX value defined by countmax register. This process then repeats periodically. Once the measurement cycle is complete, the OVF flag is set to '1' and an IRQ is triggered if the option is enabled by the IEN_OVF bit in the control register. Figure 1-11 shows the operation of the counter in variable duty-cycle trigger generation mode.

The logic high duration of the TX_TRIGGER signal is defined by the THRESHOLD value in threshold register and the period of the TX_TRIGGER signal is defined by the COUNTMAX value in the countmax register and they can be calculated as below:

$$TX_TRIGGER_{HighTime} \text{ (seconds)} = THRESHOLD \times t_{CLK} \times \text{prescaler}$$

$$TX_TRIGGER_{period} \text{ (seconds)} = COUNTMAX \times t_{CLK} \times \text{prescaler}$$

$$\text{where: } t_{CLK} = \frac{1}{f_{CLK(Hz)}}$$

In the special cases where the COUNTMAX value is equal to zero or one, the timer does not run and will reset to 0. The result register reads 0 and the TX_TRIGGER output is constantly low. Also if THRESHOLD value is equal or greater than COUNTMAX, the output TX_TRIGGER is permanently set high. In Figure 1-11, it is assumed the interrupt generation is set to '1' for IEN_OVF and set to '0' for IEN_DVL in the control register settings.

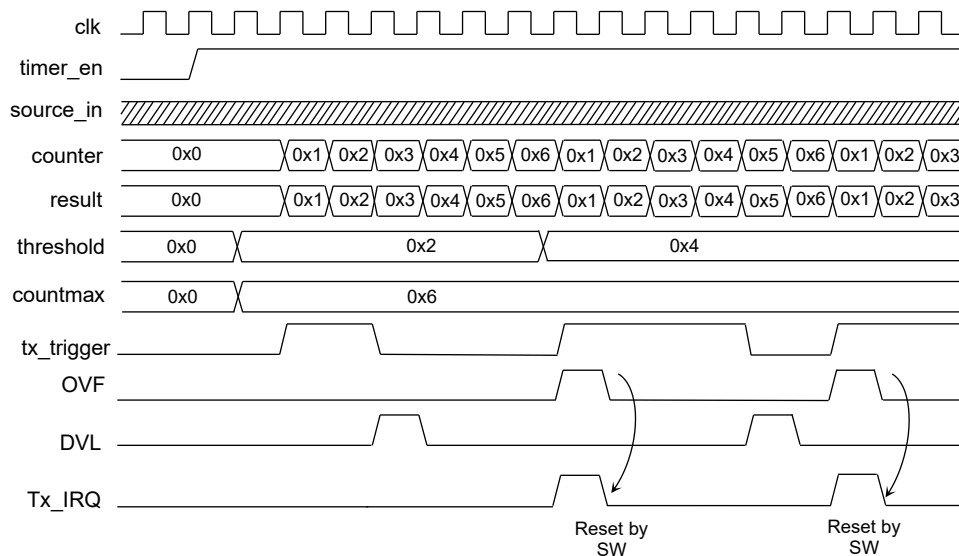


Figure 1-11: Counter's operation in Variable Duty-Cycle Trigger Generation (VDCTG)

1.5 Status management

Each timer channel has its dedicated status register to report its state. The status and result registers can be polled by software to understand the current state of the timer channel.

The status provides information about the timer operation such as: measurement completion, overflow events and data overrun events.

The bit fields in the status register are reset if the timer channel is disabled via the `TIMER_EN` bit in the control register, if the user software writes '1' to that corresponding bit field, or when the configuration of the control register is changed.

1.5.1 Data valid (DVL)

The DVL bit's meaning depends on the selected mode of operation of the timer channel:

- In measurements modes (`TMODE = 0x0 TO 0x3`), this bit indicates that the current measurement has been completed and the result is valid and available in the result register.
- In counter modes (`TMODE = 0x4 TO 0x6`) and trigger generation modes (`TMODE = 0x7 AND 0x9`), this bit indicates that the counter has reached the programmed threshold value. The DVL operation is shown in Figure 1-6, Figure 1-7, Figure 1-8, Figure 1-9 and Figure 1-11.
- In start/stop timer mode (`TMODE = 0x8`), this bit is not used and is always kept as 0.
-

1.5.2 Overflow (OVF)

The OVF bit's meaning depends on the selected mode of operation of the timer channel:

- In measurement modes (`TMODE = 0x0 TO 0x3`), this bit indicates if the counter value is greater than the maximum possible number (`0xFFFF`). Figure 1-12 shows an overflow scenario in high-time measurement mode (`TMODE = 0x0`).
- In counter modes (`TMODE = 0x4 TO 0x6`) and `FDCTG` (`TMODE = 0x7`), this bit is not used and is always kept at '0'. The counter runs up to the programmed threshold value which is also a 16-bit number. Therefore, an overflow event will never occur.
- In start/stop timer mode (`TMODE = 0x8`), this bit indicates when the timer-counter has reached its maximum value (`0xFFFF`) and rolled over to `0x0000`. This scenario is shown in Figure 1-10
- In variable duty-cycle trigger generation mode (`TMODE = 0x9`), this bit indicates that the counter value has reached to the `COUNTMAX` value as shown in Figure 1-11.

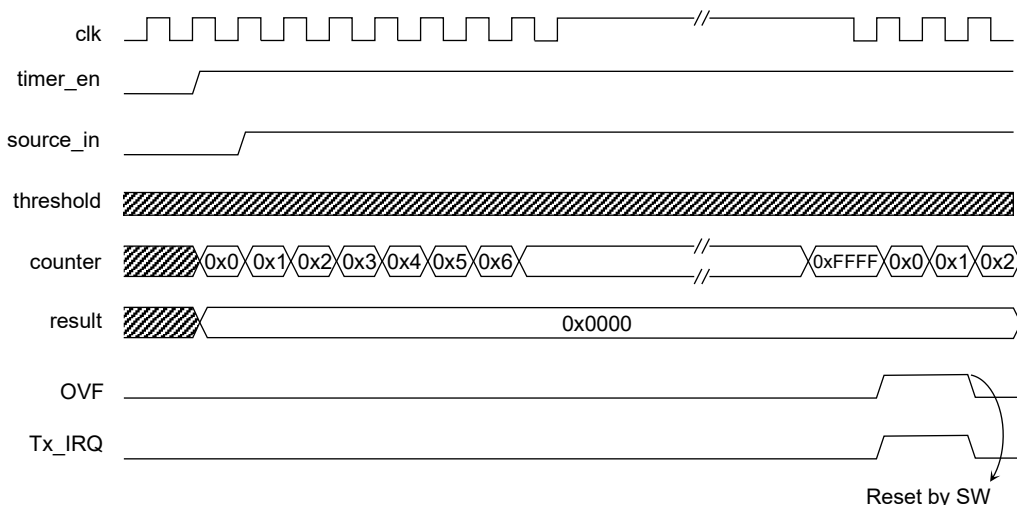


Figure 1-12: Overflow event in high-time period measurement mode (`TMODE = 0x0`)

1.5.3 Data overrun (DOR)

The DOR bit is an additional bit that is used along with the DVL bit when a new measurement has been completed and the DVL is already set to 1. This will inform the user that a measurement has been missed.

- The DOR bit is used in measurement modes (TMODE = 0x0 TO 0x3), event counter modes (TMODE = 0x4 TO 0x6), and trigger generation timer modes (TMODE = 0x7 AND 0x9). It indicates that a consecutive measurement has been completed or the threshold value has been reached again when the DVL has not been cleared yet.
- The DOR bit is not used in the start/stop timer mode (TMODE = 0x8), as the DVL is not used.

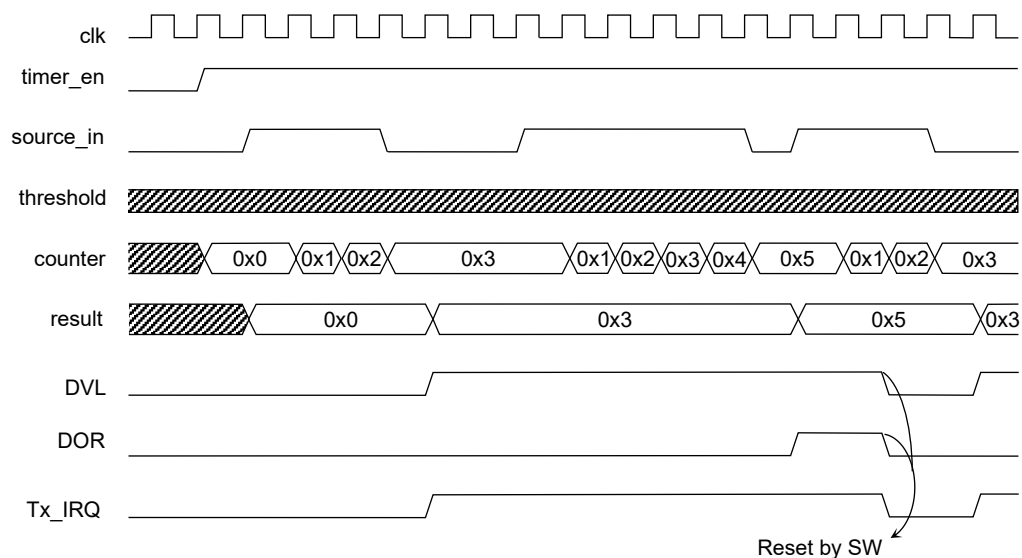


Figure 1-13: Data overrun event in high-time measurement mode (TMODE = 0x0)

1.6 Interrupt management

The GTU hosts 8 independent timer channels that can be used simultaneously. Each timer channel has its corresponding set of registers used to configure and monitor the timer's status. The result and status registers of a timer channel can be polled by SW to check the timer's current state.

Alternatively, if IRQ are preferred, each timer channel has its own independent interrupt request output Tx_IRQ that is connected to the host's NVIC. Each interrupt request can be generated individually for each timer channel.

To enable interrupt generation for a given timer channel, the feature must be enabled via the IEN_OVF and IEN_DVL bits in the corresponding control register. By default, all interrupt generation features are disabled.

An active high interrupt request Tx_IRQ is generated in one of the following cases as shown in Figure 1-14:

1. **Overflow event:** An interrupt request is generated when an overflow event occurs asserting the OVF bit in the status register to '1' and the interrupt generation is enabled by setting the IEN_OVF bit in the control register to '1'.
2. **Data valid event:** An interrupt request is generated when data is available in the result register to be read or the threshold value has been reached. This event may be signalled either by the assertion of DVL or DOR bits in the status register. The interrupt generation must be enabled via the IEN_DVL bit in the corresponding control register.

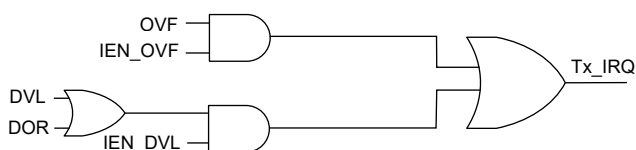


Figure 1-14: Interrupt request generation logic of a single timer channel

To reset an interrupt request generated by a timer channel, the user software shall clear OVF or DVL and DOR bits by writing '1' to their bit field positions in the corresponding status register. In doing so, the Tx_IRQ signal will also be cleared.

The GTU generates up to 8 different independent interrupt requests to the host NVIC as shown in Figure 1-15.

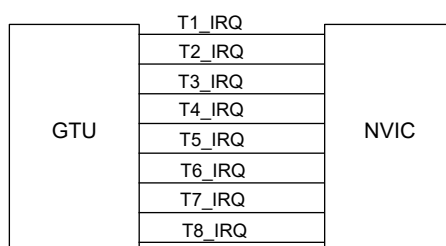


Figure 1-15: GTU interrupt requests

1.7 Register Set

The registers are accessible by software via the APB bus interface. Registers are word aligned. Reserved and unimplemented bits are read as zero. Each timer channel has six 16-bit registers to configure its functionality; each channel features a control, threshold, startstop, countmax, result and status register.

The register map of the GTU consists of 48 registers. Table 1 summarizes the register addresses, access types and reset values. The starting address of the GTU module is 0x40009100.

Base Address: 0x40009100

Table 1: Register map of the GTU

Register Name	Access Type	Address Offset	Reset Value
control1	R/W	0x000	0x0000
threshold1	R/W	0x004	0x0000
startstop1	R/W	0x008	0x0000
countmax1	R/W	0x00C	0xFFFF
result1	R	0x010	0x0000
status1	R/1C	0x014	0x0000
control2	R/W	0x100	0x0000
threshold2	R/W	0x104	0x0000
startstop2	R/W	0x108	0x0000
countmax2	R/W	0x10C	0xFFFF
result2	R	0x110	0x0000
status2	R/1C	0x114	0x0000
control3	R/W	0x200	0x0000
threshold3	R/W	0x204	0x0000
startstop3	R/W	0x208	0x0000
countmax3	R/W	0x20C	0xFFFF
result3	R	0x210	0x0000
status3	R/1C	0x214	0x0000
control4	R/W	0x300	0x0000
threshold4	R/W	0x304	0x0000
startstop4	R/W	0x308	0x0000
countmax4	R/W	0x30C	0xFFFF
result4	R	0x310	0x0000
status4	R/1C	0x314	0x0000
control5	R/W	0x400	0x0000
threshold5	R/W	0x404	0x0000
startstop5	R/W	0x408	0x0000
countmax5	R/W	0x40C	0xFFFF
result5	R	0x410	0x0000
status5	R/1C	0x414	0x0000
control6	R/W	0x500	0x0000
threshold6	R/W	0x504	0x0000
startstop6	R/W	0x508	0x0000
countmax6	R/W	0x50C	0xFFFF
result6	R	0x510	0x0000
status6	R/1C	0x514	0x0000
control7	R/W	0x600	0x0000
threshold7	R/W	0x604	0x0000
startstop7	R/W	0x608	0x0000
countmax7	R/W	0x60C	0xFFFF
result7	R	0x610	0x0000
status7	R/1C	0x614	0x0000
control8	R/W	0x700	0x0000
threshold8	R/W	0x704	0x0000
startstop8	R/W	0x708	0x0000
countmax8	R/W	0x70C	0xFFFF
result8	R	0x710	0x0000
status8	R/1C	0x714	0x0000

Control

Reset value = 0x0000

Register Address - control1: 0x40009100
control2: 0x40009200
control3: 0x40009300
control4: 0x40009400
control5: 0x40009500
control6: 0x40009600
control7: 0x40009700
control8: 0x40009800

This register is used to set input source to timer unit, clock prescaler, mode of operation, interrupt enable signals and enable the corresponding timer module.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
TIMER_EN	IEN_DVL	IEN_OVF	TMODE3	TMODE2	TMODE1	TMODE0	TCLK4
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
TCLK3	TCLK2	TCLK1	TCLK0	SRC3	SRC2	SRC1	SRC0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
SRC[3:0]^(1,6)	3:0	Configures the selected input source signal to be used by the timer-counter. 0000 _b : GPI[0] 1000 _b : sci0_rx0 0001 _b : GPI[1] 1001 _b : sci1_rx1 0010 _b : GPI[2] 1010 _b : pwm_centre 0011 _b : GPI[3] 1011 _b : bemf_comp_tacho 0100 _b : GPI[4] 1100 _b : lcrx_sync 0101 _b : GPI[5] 1101 _b : T5_TRIGGER 0110 _b : GPI[6] 1110 _b : T6_TRIGGER 0111 _b : GPI[7] 1111 _b : T7_TRIGGER
TCLK[4:0]⁽⁶⁾	8:4	Configures the timer-counter's clock pre-scaler. This will affect the timer-counter's source sample rate and the timer's increment rate. 00000 _b : clock pre-scaler = 1:1 01001 _b : clock pre-scaler = 1:512 00001 _b : clock pre-scaler = 1:2 01010 _b : clock pre-scaler = 1:1024 00010 _b : clock pre-scaler = 1:4 01011 _b : clock pre-scaler = 1:2048 00011 _b : clock pre-scaler = 1:8 01100 _b : clock pre-scaler = 1:4096 00100 _b : clock pre-scaler = 1:16 01101 _b : clock pre-scaler = 1:8192 00101 _b : clock pre-scaler = 1:32 01110 _b : clock pre-scaler = 1:16384 00110 _b : clock pre-scaler = 1:64 01111 _b : clock pre-scaler = 1:32768 00111 _b : clock pre-scaler = 1:128 10000 _b : clock pre-scaler = 1:65536 01000 _b : clock pre-scaler = 1:256 10001 _b -11111 _b : reserved
TMODE[3:0]⁽⁶⁾	12:9	Configures the mode of operation of the timer-counter. 0000 _b : Counter mode – High time measurement • Measures the high time on the selected input source signal ^(2,3) . 0001 _b : Counter mode – Low time measurement • Measures the low time on the selected input source signal ^(2,3) . 0010 _b : Counter mode – Period measurement, rising to rising edge

- Measures the period between two consecutive rising edges^(2,3).

0011_b: **Counter mode – Period measurement, falling to falling edge**

- Measures the period between two consecutive falling edges^(2,3).

0100_b: **Counter mode – Event counter, rising edges**

- Counts the number of detected rising edges on the selected input source signal until it reaches the programmable threshold stored in the threshold register^(3,4).

0101_b: **Counter mode – Event counter, falling edges**

- Counts the number of detected falling edges on the selected input source signal until it reaches the programmable threshold stored in the threshold register^(3,4).

0110_b: **Counter mode – Event counter, rising or falling edges**

- Counts the number of detected rising OR falling edges on the selected input source signal until it reaches the programmable threshold stored in the threshold register^(3,4).

0111_b: **Fixed Duty-Cycle Trigger Generation (FDCTG)**

- The timer increments based on the selected clock settings defined by TCLK without making use of the input source signal⁽³⁾.
- Generates a 50% duty cycle output signal Tx_TRIGGER whose period is defined by the value stored in the threshold register.

1000_b: **Start-Stop timer mode**

- The timer increments based on the selected clock settings defined by TCLK without making use of the input source signal⁽⁴⁾.
- The timer starts or stops counting by programming the RUN_NSTOP bit in the StartStop register.
- The timer counter rolls over to 0 when it overflows, setting the OVF flag in the status register on a roll-over event.

1001_b: **Variable Duty-Cycle Trigger Generation (VDCTG)**

- The timer increments based on the selected clock settings defined by TCLK without making use of the input source signal⁽³⁾.
- Generates a variable duty cycle trigger output Tx_TRIGGER. The period of the trigger signal is defined by the countmax register. The duty cycle of the trigger signal is defined by the threshold register.

1001-1111: Reserved.

IEN_OVF	13	Interrupt request (Tx_IRQ) enable bit when an overflow is detected. Request is triggered when OVF bit in the status register is set to '1'. 0: Interrupt request disabled 1: Interrupt request enabled
IEN_DVL	14	Interrupt request (Tx_IRQ) enable bit when data is valid, or when data overrun occurs. Request is triggered when DVL or DOR bits in the status register are set to '1'. 0: Interrupt request disabled 1: Interrupt request enabled
TIMER_EN ⁽⁵⁾	15	Master enable control bit for the timer-counter. 0: Timer-counter disabled 1: Timer-counter enabled

Notes: (1) Changing the SRC field will select a new input source. Edge detection logic will be disabled temporarily for one clock cycle right after source switching to avoid any false edge detection due to source switching.

- (2) Measurement can be read from the result register when the data is valid. Data validation is indicated by the DVL bit in the status register.
- (3) The counter-timer is restarted automatically once the measurement is done, threshold is reached, or an overflow is detected. It will automatically continue counting or measuring the new signal in the next clock cycle.
- (4) The result register shows the current counter value during event counter mode. Will be automatically reset when the threshold is reached, and a new event is detected.
- (5) Disabling the timer-counter clears all internal timer-counter logic and resets the result and status registers.
- (6) Writing new data to TMODE, TCLK and SRC fields leads to new configuration settings and will also clear the entire timer-counter logic as follows:
 - a. The current timer-counter value will be reset to zero.
 - b. The result and status registers will be reset
 - c. The trigger output signal (TMODE = 7 and 9) will be reset
 - d. The selected type of measurement or timer mode of operation will be restarted

Threshold

Reset value = 0x0000

Register Address - threshold1: 0x40009104
 threshold2: 0x40009204
 threshold3: 0x40009304
 threshold4: 0x40009404
 threshold5: 0x40009504
 threshold6: 0x40009604
 threshold7: 0x40009704
 threshold8: 0x40009804

This register is used to set the threshold for event counter modes and trigger generation modes. This register does not have any impact in measurement modes and start-stop mode.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
THRESHOLD [15:8]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
THRESHOLD [7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
THRESHOLD[15:0]	15:0	Unsigned 16-bit number specifying the threshold value used by the corresponding timer-counter. The threshold value is used in each mode as follows:

- **Measurement modes (TMODE = 0-3):** The threshold register is not used in these modes. The content of this register is irrelevant.
- **Event Counter modes (TMODE = 4-6):** When the counter value reaches the threshold value, the DVL flag in the status register is set to '1' to indicate that a measurement cycle is complete, and the threshold is reached. Meaning that the number of events detected was equal to the programmed threshold value. The counter will automatically restart and count the new events starting from 1.
In the special case where the threshold value is 0, no event counting will take place. The status register, event counter and result register are all reset to 0.
- **Fixed Duty-Cycle Trigger Generation (FDCTG):** In this mode, the threshold register specifies the period of the generated trigger output signal in increments of the selected timer increment rate TCLK.

$$tx_trigger_{period} \text{ (seconds)} = THRESHOLD \times t_{CLK} \times 2^{prescalar} \quad (1)$$

$$\text{where: } t_{CLK} = \frac{1}{f_{CLK}}$$

In the special case where the threshold value is 0 or 1, no trigger generation takes place. The trigger output remains low. The DVL bit and the result register are reset to 0.

- **Start-Stop timer mode (TMODE = 8):** The threshold register is not used in this mode. The content of this register is irrelevant.

- **Variable Duty-Cycle Trigger Generation (VDCTG):** In this mode, the threshold register specifies the logic high time duration of the generated trigger output (Tx_Trigger) in increments of the selected timer increment rate t_{CLK} .

$$tx_trigger_{LogicHigh} \text{ (seconds)} = \text{THRESHOLD} \times t_{CLK} \times 2^{prescalar} \quad (2)$$

$$\text{where: } t_{CLK} = \frac{1}{f_{CLK}}$$

StartStop

Reset value = 0x0000

Register Address - StartStop1: 0x40009108
 StartStop2: 0x40009208
 StartStop3: 0x40009308
 StartStop4: 0x40009408
 StartStop5: 0x40009508
 StartStop6: 0x40009608
 StartStop7: 0x40009708
 StartStop8: 0x40009808

This register is only used in start-stop mode only to initiate or stop the counter in the timer unit.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	-	-	RUN_NSTOP
U	U	U	U	U	U	U	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
RUN_NSTOP	0	Run or stop bit. This bit is used by the user software to control the operation of the timer channel on-the-fly when the timer is configured to operate in Start-Stop timer mode (TMODE = 8). When this bit is set to 1, the timer channel starts incrementing at the selected clock rate configured by TCLK in the control register. When this bit is set to 0 the timer freezes and keeps the current state/count until enabled again by setting the RUN_NSTOP bit to 1. When re-enabled, the timer resumes counting from the current frozen value. On reaching the maximum value (0xFFFF), the OVF flag in the status register is set and the timer rolls over to 0 continuing its operation. In all other modes of operation the contents of this register have no effect on the timer's operation. 0: Freeze 1: Run

CountMax

Reset value = 0xFFFF

Register Address - CountMax1: 0x4000910C
 CountMax2: 0x4000920C
 CountMax3: 0x4000930C
 CountMax4: 0x4000940C
 CountMax5: 0x4000950C
 CountMax6: 0x4000960C
 CountMax7: 0x4000970C
 CountMax8: 0x4000980C

This register is used to set the maximum counter value to which the timer counts for the VDCTG mode only.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
COUNTMAX [15:8]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
COUNTMAX[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
COUNTMAX[15:0]	15:0	Unsigned 16-bit number specifying the maximum value to which the counter counts. This parameter is only used in VDCTG mode.

- **Variable Duty-Cycle Trigger Generation (TMODE=9):** In this mode, the COUNTMAX defines the maximum value to which the timer counts before it resets. This register specifies the period of the generated trigger output, Tx_Trigger, in increments of the selected timer increment rate TCLK.

$$tx_trigger_period \text{ (seconds)} = COUNTMAX \times t_{CLK} \times 2^{prescalar} \quad (3)$$

$$\text{where: } t_{CLK} = \frac{1}{f_{CLK}}$$

In the special case where the COUNTMAX value is 0 or 1, no trigger generation takes places. The trigger output, Tx_Trigger, remains low. The DVL bit and the result register are reset to 0. Also, if COUNTMAX value is smaller than or equal to THRESHOLD programmed by threshold register, then the trigger output, Tx_Trigger, is permanently set high.

Result

Reset value = 0x0000

Register Address -
 result1: 0x40009110
 result2: 0x40009210
 result3: 0x40009310
 result4: 0x40009410
 result5: 0x40009510
 result6: 0x40009610
 result7: 0x40009710
 result8: 0x40009810

In measurement mode, this register is updated with high time, low time or period measurement upon completion of desired selection. In event trigger mode, this register is incremented each time the desired event (rising, falling, rising or falling) occurs until the desired threshold is reached and then reset. In FDCTG, VDCTG and start-stop modes, this register is incremented at every new clock cycle (f_{clk}).

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
RESULT[15:8]							
R	R	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
RESULT[7:0]							
R	R	R	R	R	R	R	R

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
RESULT[15:0]	15:0	Unsigned 16-bit number representing the value of the completed measurement or the counter value. The result register's meaning is different in each mode of operation:

- **Measurement modes (TMODE = 0-3):** In these modes, the result register is updated upon the completion of the desired measurement. Once the measurement is completed, the DVL bit in the status register is set to '1', indicating that data is valid and available, and the measured value is stored in the result register. The stored value represents the measured duration as follows:

$$\text{Duration (seconds)} = \text{RESULT} \times t_{CLK} \times \text{prescaler} \quad (4)$$

$$\text{where: } t_{CLK} = \frac{1}{f_{CLK}}$$

- **Event Counter modes (TMODE = 4-6):** In these modes, the result register is continuously updated with the current number of detected events on the selected source signal. When the count value reaches the defined threshold value and DVL bit in the status register is set to '1' indicating the end of the measurement cycle. The result register is automatically reset and is updated with the new counter value of the new measurement cycle.
- **Fixed Duty-Cycle Trigger Generation (FDCTG) (TMODE = 7):** In this mode, the result register is continuously updated with the timer value. When the timer value reaches the defined THRESHOLD value and DVL bit in the status register is set to '1', indicating the end of the cycle. the result register is automatically reset and updated with the new timer increment of the new trigger generation cycle.

- **Start-Stop timer mode (TMODE = 8):** In this mode, the result register is updated continuously with the current value of the timer. When the timer reaches its maximum value (0xFFFF), the OVF bit in the status register is set to 1, indicating an overflow event. The timer rolls-over and continues counting from 0.
- **Variable Duty-Cycle Trigger Generation (VDCTG) (TMODE = 9):** In this mode, the result register is continuously updated with the timer value. When the timer value reaches the defined COUNTMAX value and OVF bit in the status register is set to '1', indicating the end of the cycle, the result register is automatically reset and updated with the new timer increment of the new trigger generation cycle.

Status

Reset value = 0x0000

Register Address - status1: 0x40009114
 status2: 0x40009214
 status3: 0x40009314
 status4: 0x40009414
 status5: 0x40009514
 status6: 0x40009614
 status7: 0x40009714
 status8: 0x40009814

This register reports the status information in the timer module. The status bits can only be cleared by writing '1' to the corresponding bit.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	-	DOR	DVL	OVF
U	U	U	U	U	R/1C	R/1C	R/1C

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
OVF	0	Overflow flag. This flag is set differently in each mode of operation: • Measurement modes (TMODE = 0-3): In these modes, the flag is set when the measured period is greater than the maximum possible counter number (0xFFFF). To increase the maximum measurable duration, increase pre-scaler value from the TCLK bits in the control register to slow down the input sample rate⁽¹⁾. • Event Counter modes (TMODE = 4-6): In these modes, the OVF flag is not used and is kept as '0'. As the event counter runs up to the programmed Threshold value, which is a 16-bit value, and then rolls back to 1. Therefore, an overflow will never happen, and this bit becomes obsolete. • Fixed Duty-Cycle Trigger Generation (FDCTG) (TMODE = 7): In this mode, the OVF flag is not used and is kept as '0'. The timer runs up to the programmed Threshold value, which is a 16-bit value, and then rolls back to 1. Therefore, an overflow will never happen, and this bit becomes obsolete. • Start-Stop timer mode (TMODE = 8): In this mode, the OVF flag is set when the timer-counter has reached the maximum value (0xFFFF) and rolls over to 0. • Variable Duty-Cycle Trigger Generation (VDCTG) (TMODE = 9): In this mode, the OVF flag is set when the timer count reaches to a value programmed by the countmax register.

The OVF flag is reset in any of the following conditions:

- TIMER_EN bit in the control register is set to '0'.
- Writing '1' to this bit location.
- Configuration of control register is changed.

		0: No overflow has occurred. 1: Overflow event occurred/
DVL	1	Data valid flag. This flag is set differently in each mode of operation: • Measurement modes (TMODE = 0-3): In these modes, the flag is set when a valid measurement is complete. The measured data is available in the result register. • Event Counter modes (TMODE = 4-6): In these modes, the flag is set when the counter reaches the programmed threshold value. • Fixed Duty-Cycle Trigger Generation (FDCTG) (TMODE = 7): In this mode, the flag is set when the timer reaches the programmed threshold value. • Start-Stop timer mode (TMODE = 8): In this mode, the DVL flag is not used and is always set to '0'. • Variable Duty-Cycle Trigger Generation (FDCTG) (TMODE = 7): In this mode, the flag is set when the timer reaches the programmed threshold value. The DVL flag is reset in any of the following conditions: • TIMER_EN bit in the control register is set to '0'. • Writing '1' to this bit location. • Configuration of control register is changed.
DOR	2	Data overrun flag. This flag is an addition to the DVL flag. This bit is set whenever a new measurement is completed but the DVL flag is set to '1'⁽²⁾. This bit can be used to detect when the user software or ISR has missed a measurement. The DOR flag is reset in any of the following conditions: • TIMER_EN bit in the control register is set to '0'. • Writing '1' to this bit location. • Configuration of control register is changed. 0: No overrun detected 1: Overrun detected Note: (1) Slowing down input sample rate extends the maximum measurable period but decreases the resolution as a trade-off. (2) This is done by copying the current content value of DVL to DOR bit whenever DVL is set to 1.

Revision Control:

Feb/17/2021	Initial Version
Feb/26/2021	Correct Register Address naming For Countmax
Nov/11/2021	Updated Logo

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